Identification of parent of origin specific allelic expression in outbreeding *Arabidopsis arenosa* species unravel paternal bias in protein degradation pathways

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Keywords:

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Abstract

In plants, genomic imprinting mainly takes place in the triploid endosperm and has been extensively researched in the inbreeding model species *A. thaliana*. Genomic imprinting refers to the epigenetic phenomenon of parent-of-origin allele specific expression, observed mainly in mammals and flowering plants. Although parent-of-origin dependent expression is well studied in the hermaphrodite species *A. thaliana*, genomic imprinting is less investigated in outcrossing species. High-throughput analysis of genomic imprinting in outbreeding species is hampered by the lack of reference genomes and available sequenced ecotype accessions to allow for the detection of allele specific expression patterns. To this end we have generated *de novo* genome assemblies of the outbreeder *A. arenosa* and analyzed imprinting in *A. arenosa* using re-sequenced *A. arenosa* accessions in outbreeding species, allele variation may not be captured in the references used for mapping and may therefore limit the analysis to loci with less variation. In this report we therefore compare traditional reference based analysis (IRP) with a k-mer based approach (KIRP) that allows detection of parental specific reads independent of a reference transcriptome for the different accessions. We report that KIRP detection of imprinted genes surpasses traditional methods of imprinted gene detection. A total of 71 reciprocal paternally biased genes were identified and showed enrichment for proteins with ubiquitin protein transferase and ligase activity. Strikingly, maternally expressed imprinted genes were enriched for protein pathways directly counteracting paternally enriched genes, suggesting a unique role of genomic imprinting in gene regulation at the protein level in outbreeding *A. arenosa*.

Introduction

Genomic imprinting refers to the epigenetic phenomenon of parent-of-origin allele specific expression, observed mainly in mammals and flowering plants (Feil and Berger 2007). In plants, genomic imprinting mainly takes place in the triploid endosperm and has been extensively researched in the inbreeding model species *A. thaliana* (Grossniklaus et al. 1998; Jullien and Berger 2010; Gehring et al. 2011; Hsieh et al. 2011; Wolff et al. 2011; Pignatta et al. 2014; Hornslien et al. 2019; León et al. 2019; van Ekelenburg et al. 2023). Endosperm development depends on a correct ratio of parental genomes, and it is suggested that differences in genomic imprinting programs may be responsible for the evolution of sexual incompatibility in crosses between divergent individuals (Haig and Westoby 1988, 1991; Schatlowski and Köhler 2012). Imprinting in the endosperm and parent-of-origin dependent expression of genes is due to deposition of epigenetic marks in the male and female gametes. Canonically, this process involves partial or full silencing of one of the parental alleles and is associated with both DNA methylation and histone modification (Jullien et al. 2006; Gehring and Satyaki 2017). Furthermore, the involvement of small RNAs through the RNA-directed DNA methylation (RdDM) pathway resulting in de novo DNA methylation has been demonstrated (Vu et al. 2013; Gehring and Satyaki 2017; Hornslien et al. 2019; Batista and Kohler 2020).

Although parent-of-origin dependent expression is well studied in the hermaphrodite species *A. thaliana*, genomic imprinting is less investigated in outcrossing species. Within the genus *Arabidopsis*, an imprinting study in *A. lyrata* has pointed towards conservation of imprinted genes between *A. thaliana* and *A. lyrata* (Klosinska et al. 2016; Erdmann et al. 2017), but conservation of imprinting in plants is disputed (Waters et al. 2013; Hatorangan et al. 2016). High-throughput analysis of genomic imprinting in outbreeding species is hampered by the lack of reference genomes and available sequenced ecotype accessions to allow for the detection of allele specific expression patterns. To this end we have generated *de novo* genome assemblies of the outbreeder *A. arenosa* (Bramsiepe et al. 2023). Here we have analyzed imprinting in *A. arenosa* using re-sequenced *A. arenosa* accessions Pusté Pole and Strecno (Jørgensen et al. 2011; Hollister et al. 2012). In outbreeding species, allele variation may not be captured in the references used for mapping and may therefore limit the analysis to loci with less variation. In this report we therefore compare traditional reference based analysis (IRP) (Hornslien et al. 2019; van Ekelenburg et al. 2023) with a k-mer based approach (KIRP) that allows detection of parental specific reads independent of a reference transcriptome for the different accessions. We report that KIRP detection of imprinted genes surpasses traditional methods of imprinted gene detection. A total of 71 reciprocal paternally biased genes were identified, more than twice the amount detected with a traditional IRP approach. Interestingly, paternally expressed genes showed significant 10-fold enrichment for proteins with ubiquitin protein transferase and ligase

activity. Maternally expressed imprinted genes on the other hand displayed 20-fold enrichment for proteins directly involved in the COP9 signalosome pathway, directly counteracting paternally enriched ubiquitin ligases. Similar enrichment in the E3 ubiquitin ligase pathway or COP9 signalosome could not be found re-analysing 273 unique *A. thaliana* PEGs or 1141 unique *A. thaliana* MEGs (Gehring et al. 2011; Hsieh et al. 2011; Wolff et al. 2011; Pignatta et al. 2014; Hornslien et al. 2019; Picard et al. 2021; van Ekelenburg et al. 2023), suggesting a unique role of genomic imprinting in gene regulation at the protein level in outbreeding *A. arenosa*.

Results and Discussion

With the aim to identify genes with parent-of-origin dependent gene expression in an outbreeding species we used diploid *A. arenosa* (Bramsiepe et al. 2023) originating from two different Slovakian populations, Strecno and Pusté Pole (Jørgensen et al. 2011; Hollister et al. 2012) (Figure 1a). The two populations have differences in some phenotypic characteristics such as flower size, rosette, leaf shape and overall architecture (Figure 1b) and are regarded as different ecotypes. Seed development was monitored for the inter- and intra-population crosses in order to determine the time point when most embryos had reached globular stage, but not yet progressed into transition stage. Seed developmental progression in the Pusté Pole ecotype was slightly faster than in the Strecno ecotype, and the seed development progression in the inter-population crosses followed the maternal parent (Figure 1c-d, STable 1). Material for whole seed transcriptome analysis from crosses was harvested when seed development had reached the globular embryo stage for reciprocal crosses and controls (nine or ten days after pollination (DAP) for Pusté Pole and Strecno maternal crosses, respectively) (SData 1).

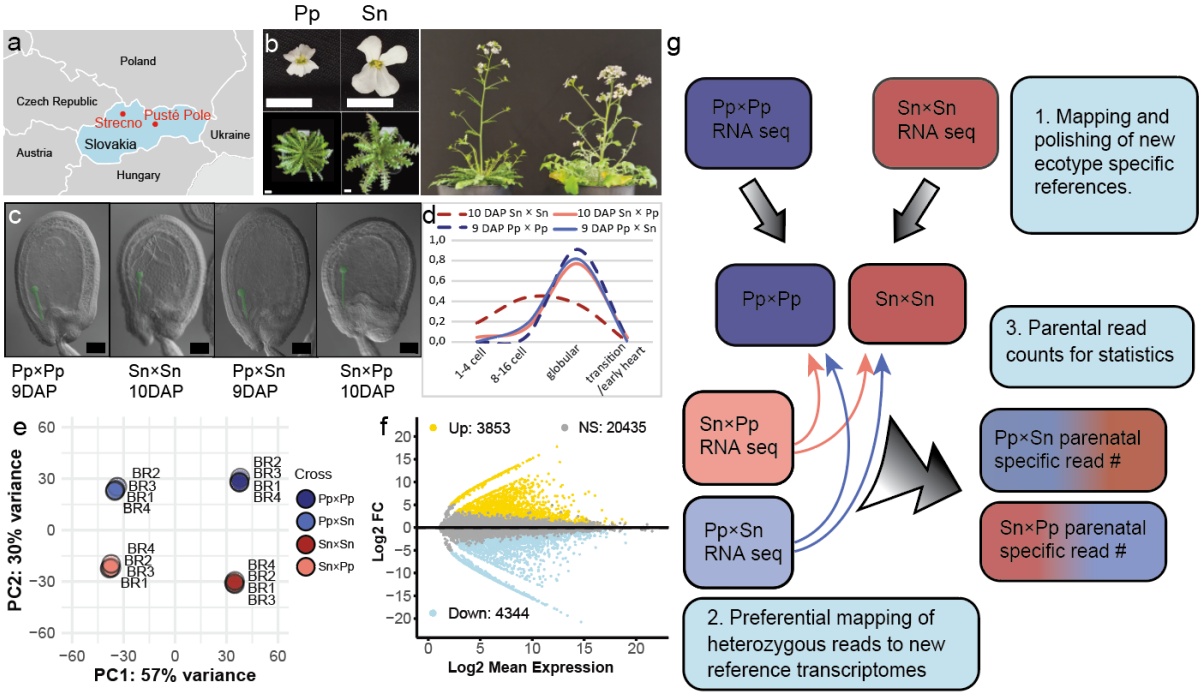

Figure 1. a) *A. arenosa* accessions from populations collected in Pusté Pole (Pp) and Strecno (Sn) in Slovakia. b) The two natural populations have distinct characteristics related to size of flower, rosettes, leaf shape and overall architecture. c-d) Monitoring of seed development shows that Pp accession develops at a slightly higher speed than Sn, and that development in inter-accession crosses follows the maternal parent. Tissue was harvested for RNA sequencing when most seeds had reached globular stage but not yet progressed into transition/early heart, 9 days after pollination (DAP) for Pp as maternal parent and 10 DAP with Sn as the maternal parent. e) Principal component analysis of mapped reads show homogeneity between samples, clustering on the maternal parent, and distinction between hybrid cross and non-hybrid cross. f) Differential expression analysis between Pp and Sn show some distinct expression differences between accessions but most genes are not significantly differentially expressed. g) Reads from intra-accession crosses were used in the Informative read Pipeline to generate and polish accession specific

transcriptomes. RNA sequencing reads from hybrid crosses were subsequently mapped to the new transcriptomes following the IRP pipeline to identify parent of origin dependent expression patterns. White scale bars: 10 mm, black scale bars: 100 µM.

RNA seq from four replicates of control and hybrid reciprocal crosses produced on average 80-100 millions of read pairs (SData 2). To aid analysis of the sequencing reads derived from different ecotypes, or from crosses between them, we initially mapped reads from homozygous crosses separately to the *de novo* assembled *A. arenosa* reference (Bramsiepe et al. 2023). Subsequently this reference transcriptome was polished by iterations of Pilon and Bowtie to generate ecotype specific transcriptomes for Pusté Pole and Strecno (Hornslien et al. 2019; van Ekelenburg et al. 2023). PCA of mapped reads to the new ecotype specific references (SData 2) showed homogeneity between replicates, and the reciprocal crosses were most similar to their maternal parent (Figure 1e, SData 3, SFigure 1). Additionally, the crosses cluster based on the nature of cross as non-hybrid or hybrid. Differential gene expression analysis between ecotypes showed a substantial number of differentially expressed genes both between the controls and between the reciprocal crosses (Figure 1f, SData 4), although the majority of genes showed no significant differences.

In order to identify parent of origin allelic expression we used the informative reads protocol (IRP) (Hornslien et al. 2019; van Ekelenburg et al. 2023) (Figure 1g). Reads from heterozygous crosses are classified as maternal or paternal if it shows preferential mapping to one of the reference transcriptomes. If a read mapped to the same gene in both transcriptomes and at the same time harbored SNPs and/or indels that distinguished the map preferences for the two transcriptomes, the read was termed informative and parent of origin was determined as maternal or paternal. Differential gene expression analysis of maternal to paternal reads within each hybrid cross identifies 3439 and 39 reciprocal maternally and paternally biased genes respectively (adjusted p-value <0,01) (Figure 2a-b, SData 5 and SData 6).

Notably, using Strecno mother yielded almost twice as many maternally biased genes as the reciprocal cross and similarly using Strecno father yielded almost twice as many paternally biased genes (Figure 2b). Regardless of this, genes were required to show consistent reciprocal parental bias in the two hybrid crosses to be further assessed as truly parentally biased. The closest homolog in *A. thaliana* was identified if present (Bramsiepe et al. 2023), and their known expression pattern in sub compartments of the seed was used as a filtering tool (Belmonte et al. 2013; Schon and Nodine 2017; Hornslien et al. 2019). 32 % of genes showed evidence of seed coat enrichment or had contradicting enrichment data in different tissues and stages of seed development (SFigure 2, SData 7). 2 % and 10 % of maternally biased genes were found to be solely enriched in the suspensor or the embryo at any stage during seed development. 43% of maternally biased genes had no homolog or no available enrichment data.

Although this gene set probably contains truly imprinted genes we cannot confidently exclude maternal bias caused by other maternal tissues in the seed. 13 % of the maternally biased genes had a homolog in *A. thaliana* that was exclusively enriched in endosperm tissues at one or several stages of seed development. None of the homologs of paternally biased genes showed enrichment in any other seed compartment than the endosperm tissues (46%), although 54 % of the paternally biased genes did not have a homolog in *A. thaliana* or the homolog did not have any enrichment data (SFigure 2).

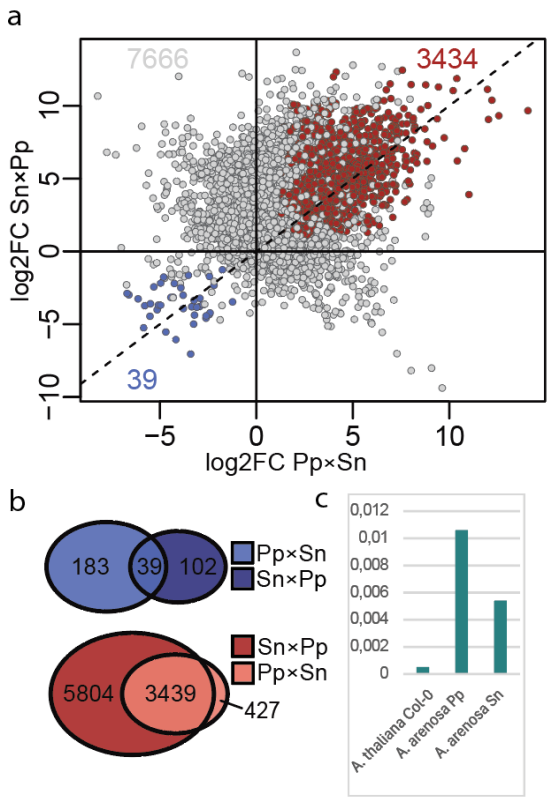

Figure 2. Differential gene expression analysis of maternal to paternal read counts reveal mapping bias to the Strecno accession. a) Maternal to paternal read counts from reciprocal inter-accession crosses Strecno (Sn) × Pusté Pole (Pp) plotted against each other show distribution of maternally and paternally biased genes. Genes that are significantly biased in the same manner in both crosses are colored (paternally biased: blue, maternally biased: red). The dotted diagonal line is added as a reference line of equality (y=x). b) venn diagram shows intersection of significantly paternally (blue) and maternally (red) biased genes in the differential gene expression analysis of reciprocal crosses. c) The ratio of heterozygous positions to all positions was calculated from the mapped RNA sequencing reads on the homozygous read set for each accession. *A. thaliana* Col-0 reads were included as a reference point for comparison.

The final sets of genes shown to have imprinted expression in the endosperm of *A. arenosa* (SData 7) is rigorous despite the observed mapping bias towards the Strecno reference. We further speculated that the mapping bias is created as an artifact from the Pilon iterations generating the final ecotype references. One possibility that may generate such obvious differences between the two ecotype references might be the level of heterozygosity within the two ecotypes. The final reference transcriptome generated for our purpose only displays one haplotype randomly fixed by Pilon. If there is a great difference in the level of heterozygosity between the two analyzed ecotypes this might create

a mapping bias towards the less heterozygous ecotype. The level of heterozygosity was calculated to be 0.01 and 0.005 for the Pusté Pole and Strecno ecotypes (Figure 2c), respectively, confirming that Pusté Pole has on average up to 10 heterozygous positions per 1000 bp in RNA sequencing data, twice the amount of Strecno. In comparison *A. thaliana* (Columbia-0) (van Ekelenburg et al. 2023) was included and found to have a heterozygosity level of 0.0005 (conferring to a half SNP per 1000 bp). Although all the ecotypes analyzed have a low degree of heterozygosity in general, the relative difference between the *A. arenosa* ecotypes is 95.05%. Similar types of map bias have been reported previously (Degner et al. 2009) and random fixation of one haplotype can lead to moderate to severe mapping bias in different scenarios (SFigure 3).

A larger variation between replicate (different maternal plant individuals but same father between replicates) also show a larger degree of variation for the Pust é Pole accession compared to the Strecno population (SFigure 1) further supporting the hypothesis that a larger degree of SNP variation is segregating in this population compared to the Strecno population.

The IRP pipeline is therefore shown to have shortcomings with respect to analysis of outbreeding populations, and although these shortcomings do not necessarily deteriorate the obtained results, they might suffer from under-reporting of reciprocally parentally biased genes. This urged us to develop a new approach to investigate parent of origin dependent gene expression independent of existing reference transcriptomes, and retaining all information on sequence variation.

Enhanced detection of parent of origin dependent gene expression with k-mers

Using the same read sets described for the Pilon-IRP approach, reads were analyzed using k-mers. Using this approach we moved away from the alignment based method to detect parental origin of transcripts. The k-mer approach (Figure 3a) allows detection of parental specific reads independent of a reference transcriptome for the different ecotypes. K-mer libraries are generated for all reads from the homozygous crosses individually, and ecotype specific k-mers are defined for k-mers that are unique to one or the other ecotype. In the second step the same procedure is done for reads from heterozygous reciprocal crosses between the ecotypes. The k-mers in the heterozygous clouds are then classified as ecotype specific, shared between ecotypes or unique to the heterozygous offspring. Ecotype specific k-mers are then classified as maternal or paternal originating k-mers, and the reads are extracted and can subsequently be mapped to a reference transcriptome with less strict mapping requirements, allowing mismatches due to SNP variation, to identify the gene of origin.

Analysing total numbers of maternal and paternal mapped reads obtained from the k-mer informative reads pipeline (KIRP), there is no longer any significant mapping bias towards the Strecno accession

(Figure 3b and S Data 8). Before any filtering steps, we detected a total of 16,674 reciprocal maternally biased genes and 71 reciprocal paternally biased genes (Figure 3c + SData 9). The distribution of genes identified in each cross direction indicates a greater degree of equality for both maternally and paternally biased genes. This finding further supports the conclusion that the mapping bias towards Strecno is now mitigated through the use of the KIRP (Figure 3d SData 10).

The comparison between the k-mer approach and the Pilon approach reveals that the k-mer method captures a nearly identical set of genes, as well as additional candidate genes (Figure 3e). Of the four paternally biased genes, not captured by k-mer approach classified as a paternally expressed gene in Pilon-IRP approach, two were only significantly paternally expressed in one direction and slightly biased but not significant in the reciprocal cross (jg19555 and jg7593). The two others did not pass the filtering step of minimum read requirement after salmon mapping (jg11459 and jg14500)? Of the 36 novel paternally biased genes detected by the k-mer approach, 20 genes did not pass the filtering criteria for minimum read count or 5-fold preference for the correct parent and was thus not analysed for parental bias in the heterozygous crosses in the Pilon-IRP approach. The remaining 16 were analyzed and all of them showed slight paternal expression in both cross directions. 12 of them were also significantly paternally biased when Strecno was used as the paternal cross partner. However, when Strecno was used as the maternal cross partner, the ratio of maternal to paternal reads was shifted to the extent that none of these 12 were found to be significantly paternally biased in the cross Strecno x Pusté Pole. The shift in ratio of maternal to paternal reads in Strecno x Pusté Pole may have been caused by the mapping bias towards Strecno observed to the Pilon generated references transcriptome.

A substantial fraction (24%) (Figure 3f, SData 11) of the identified maternally biased genes have a homolog in *A. thaliana* that has previously been identified as seed coat specific (Belmonte et al. 2013; Schon and Nodine 2017; Hornslien et al. 2019). 1882 *A. arenosa* maternally biased genes (14%) have homologs in *A. thaliana* that have any trace of endosperm enrichment (>1-fold) (Hornslien et al. 2019). Of these, 505 are shown to be at least 4-fold enriched in the endosperm. 49% of genes had either no homolog in *A. thaliana*, or did not have any compartment specific expression data in the seed (Belmonte et al. 2013).

In contrast, most paternally biased genes (52%) have previously been shown to be endosperm enriched for their respective *A. thaliana* homolog. Notably, only one gene exhibited enrichment status in seed coat tissues, but this same gene was also reported to be enriched in endosperm tissue at different stages of seed development (Belmonte et al. 2013; Schon and Nodine 2017; Hornslien et al.

2019). The remaining of paternally expressed genes (45%) either did not have any compartment specific expression data in the seed or had no homolog in *A. thaliana.*

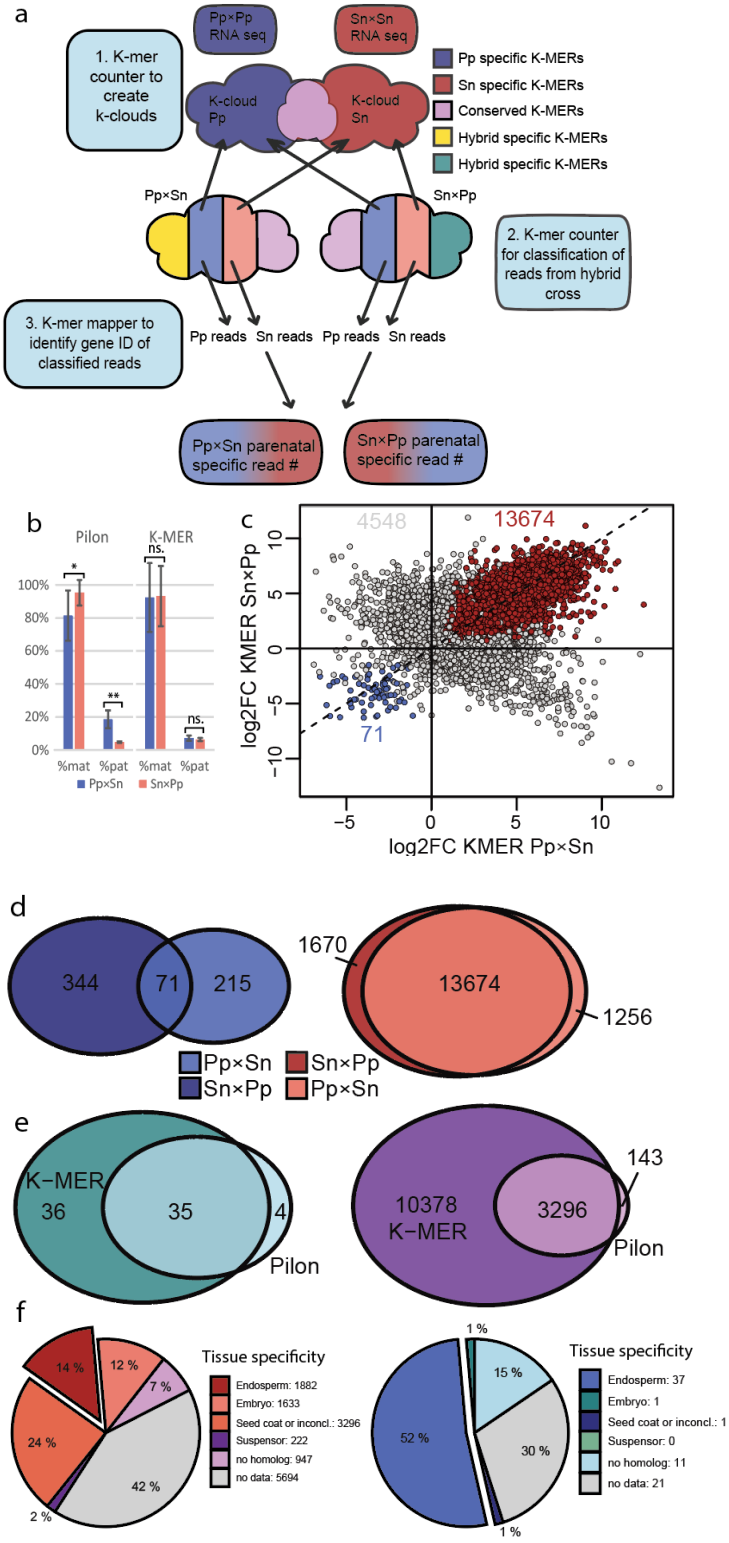

Figure 3. Detection of parental specific gene expression in inter accession crosses by K-MER approach. a) outline of the K-MER approach. b) Mapping bias towards one accession is no longer significant using the K-MER assignment of reads. c) Maternal to paternal read counts from reciprocal inter-accession crosses Strecno (Sn) × Pusté Pole (Pp) plotted against each other show distribution of maternally and paternally biased genes. Genes that are significantly biased in the same manner in both crosses are colored (paternally biased: blue, maternally biased: red) The dotted diagonal line is added as a reference line of equality (y=x) d) venn diagram shows intersection of significantly paternally (blue) and maternally (red) biased genes in the

differential gene expression analysis of reciprocal crosses. e) K-MER approach captures the same, and additional genes compared to the Pilon-IRP set. f) Maternal biased expression is in a large part shown to be seed coat expressed genes analysing published data for *A. thaliana* homologs. Paternally biased genes are confirmed to be endosperm specific analysing *A. thaliana* homologs in published data (blue).

Conservation of imprinted paternally biased genes in the *Arabidopsis* genus

Although parent of origin dependent gene expression in developing endosperm has been studied in a wide range of plant species, the number of particular genes that show a broad, conserved imprinting pattern between species has been scarce. Extensive efforts has also been put into the investigation of parent of origin dependent gene expression in the well established model organism *A. thaliana* (Gehring et al. 2011; Hsieh et al. 2011; Wolff et al. 2011; Hornslien et al. 2019; Picard et al. 2021; van Ekelenburg et al. 2023), which has resulted in excessive lists of reported imprinted genes. Some of the earlier studies have also been criticized for over-reporting the number of maternally biased genes due to interference of the maternally derived seed coat in seed samples. All paternally biased genes, and a subset of maternally biased genes showing at least a four fold enrichment in endosperm compartments (SData 12) were used further to investigate possible conservation in the *Brassicaceae* family (Figure 4) including both inbreeding and outbreeding species.

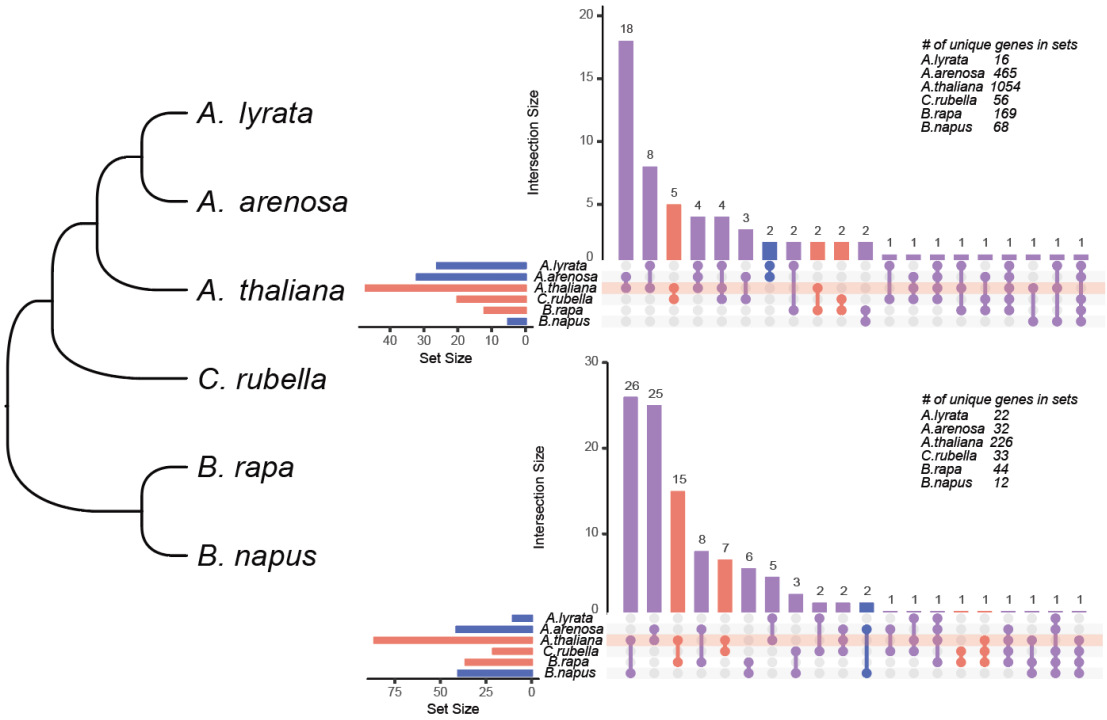

Figure 4. Conservation of parental specific expression of genes in species investigated from the *Brassicaceae* genus. Paternally biased genes (top) Maternally biased genes (bottom). blue = natural outbreeders, orange inbreeders, purple intersection of imprinted genes between different mating systems.

Parentally biased genes enriched in GO terms for protein degradation pathways

Studies have investigated the function of imprinted genes, particularly focusing on their roles in seed development. To further understand the specific contributions of parentally biased gene expression, a

Gene Ontology (GO) term analysis was conducted on the genes identified with biased expression confined to the endosperm (SData 12). This analysis aims to shed light on the functional implications of imprinting in the context of plant reproductive strategies and development. The GO enrichment analysis was performed on genes with homologs in *A. thaliana* and using tools available for GO term enrichment in *A. thaliana* as a starting point. Paternally expressed genes showed significant enrichment in four GO terms, of which almost a 10-fold enrichment was observed for protein with ubiquitin protein transferase and ligase activity (Figure 5a-top). Further on, a notable big fraction of paternally biased genes was observed to be F-box containing proteins (26%) (Figure 5b). These proteins are involved in ubiquitin protein degradation pathways. Maternally biased genes were subjected to similar GO term analysis and enrichment was found in more than 10 categories, of which six showed more than two-fold enrichment (Figure 5a-bottom). Strikingly, one GO term was shown to be enriched more than 20-fold for proteins directly involved in the COP9 signalosome pathway. The COP9 signalosome is directly linked to protein degradation by derubylation/deneddylation of Cullin RING Ligase (CRL) proteins causing destabilization of the CRL complexes.

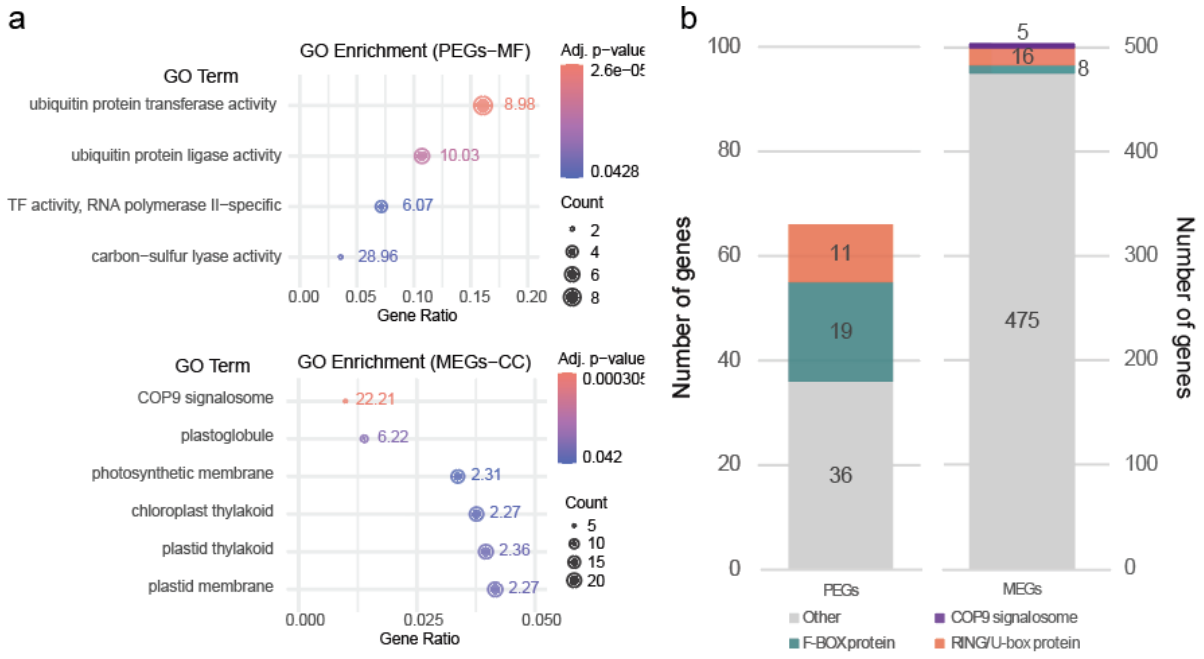

Figure 5. Gene Ontology (GO) Enrichment analysis of homologs of parentally biased genes in *A. arenosa*. a) Enrichment analysis shows significant enrichment of paternally expressed genes (PEGs) in molecular function (MF) for E3-ubiquitin ligase and transferase activity (top panel). A subset of top GO terms for maternally expressed genes (MEGs) show significant enrichment in COP9 signalosome and plastid related cellular component (CC) GO terms (bottom panel). b) Number of genes of MEGs and PEGs belonging to protein degradation pathways.

Materials and Methods

Plant material and growth conditions

Both the *A. arenosa* populations originate from Slovakia. The Pusté Pole was collected from Nízke Tatry Mountains. (N 48.8855, E 20.2408) (MJ09-4, (Jørgensen et al. 2011)) and Strecno seeds were kindly supplied by Kirsten Bomblies (SN1, (Hollister et al. 2012)). Seeds were surface sterilized by step vice washing in 70% Ethanol, 20% bleach and wash solution (0.001% Tween20) before planting on 0.5 Murashige and Skoog (MS) plates (Murashige and Skoog 1962) supplemented with 2% sucrose, and left for stratification for one week (dark, 4 °C) . Seeds were transferred to growth chambers at long-day conditions (16 h light, 18°C) for germination and approximately two weeks old seedlings were transferred to soil for vegetative growth at long-day conditions (16 h light, 18°C). Plants were vernalized for a minimum of four weeks in short-day conditions (10 h light, 8-10 ^{o}C), and flowering was initiated in long-day conditions. For phenotypic seed development analysis and RNA extraction flowers were manually crossed and developing siliques were dissected under a stereomicroscope. For RNA extraction, seeds were harvested into cooled tubes in liquid nitrogen. Each biological replicate was made up of seeds from 10 siliques of the same mother plant, four replicates for each cross. For phenotypic analysis, seeds were mounted on a microscopy slide in a clearing solution of glycerol and chloral hydrate as previously described (Grini et al. 2002). Microscopy analysis was performed using a Zeiss Axioplan Imaging2 microscope equipped with Nomarski optics.

RNA extraction and library prep

Tissue was collected to MagNA-Lyser Green Beads Tubes (Roche) and RNA was extracted using SIGMA-ALDRICH Spectrum Plant Total RNA kit. Total RNA was quality checked by Qubit broad range assay, Nanodrop and Agilent RNA6000 kit for Agilent 2100 Bioanalyzer. cDNA libraries were prepared with the Truseq stranded mRNA library prep kit from Illumina, using 500 ng total RNA as input. RNA was fragmented by incubation for 1 min at 94°C, and IDT for Illumina Truseq RNA UD indexed adapters were used. Libraries were sequenced to produce 150bp paired end reads using one and a half SP flow cell and Illumina Novaseq6000 instrument.

Image analysis

Images were processed and assembled using Adobe Photoshop and Adobe Illustrator 2023 (Adobe Systems Incorporated, San Jose, USA).

Bioinformatic analysis

Reads were trimmed with Trim_Galore v0.6.2 (Babraham Bioinformatics) (Krueger 2015). All samples were subjected to Differential expression analysis using DeSeq2 for PCA analysis. To allow ecotype specific mapping for detection of parental specific expression the informative read pipeline (IRP) was followed (Hornslien et al. 2019; van Ekelenburg et al. 2023). Reads from intra ecotype crosses were used to build an ecotype specific reference using *A. arenosa* genome (Bramsiepe et al. 2023) as the initial substrate. Only the CDS, in addition to 150 bp upstream and downstream of CDS (if not conflicting with adjacent gene or end of contig) to capture possible UTRs to maximize read mapping, was used as mapping substrate for the RNA seq reads. Bowtie2 (Langmead and Salzberg 2012) version 2.4.5 (settings: no-unal-no-mixed-no-discordant-sensitive-k1-end-to-end) and Pilon (Walker et al. 2014) version 1.23 (options-x all-changes) was used in 13 iterations steps with the same settings as previously reported for IRP (Hornslien et al. 2019; van Ekelenburg et al. 2023). Reads from each interecotype crosses were allowed to map to the two new ecotype specific references separately and informative reads were extracted. An informative read was required to have mapped concordantly with its pair (i.e. proper orientation and approximate spacing for the read pair), mapped to the same gene in both strains, both mapQ scores $ 5, and either one strain's alignment spanning InDels while the other is InDel free or both alignments InDel free but one spanning fewer mismatches (i.e. SNPs) (van Ekelenburg et al. 2023). The informative reads were categorized as maternal and paternal according to preferred mapping to one of the ecotype specific references over the other. Maternal and paternal read counts were normalized using predetermined normalization factors to correct for sequencing library sizes (van Ekelenburg et al. 2023) before they were subjected to differential gene expression analysis using the Limma package in R (Ritchie et al. 2015)

Heterozygosity was measured on the homozygous reads from Pusté Pole and Strecno ecotypes mapped to *A. arenosa* CDS (Bramsiepe et al. 2023). A comparable readset from *A. thaliana* was included (van Ekelenburg et al. 2023) and mapped to *A. thaliana* CDS (TAIR10). Heterozygous positions were counted using BCFtools v1.15.1 (Danecek et al. 2021) and the level of heterozygosity was estimated as the ratio of heterozygous sites to all sites.

Gene Ontology and Enrichment analysis was performed using Cluster profiler (Xu et al. 2024).

The K-MER Informative Read Pipeline (KIRP) was designed to be kmer-based and alignment-free. In KIRP, the parental assignment is performed by kmer analysis of RNAseq reads from two parents and their cross. Meryl software (unpublished, described in Rhie 2020 (Rhie et al. 2020), available on GitHub) was used for kmer counting and kmer content comparison. Salmon software (Patro et al. 2017) was

used to map RNAseq read pairs to transcripts for gene ID assignment; Salmon is a kmer-based, alignment-free read mapper. The limma software (Ritchie et al. 2015) was used to measure statistical significance.

Step 1. Label cross read pairs by parent or origin: The process below was repeated independently for each biological replicate (BR1 though BR4) with parental genotypes PpxPp or SnxSn and cross genotypes PpxSn or SnxPp (maternal x paternal). The kmer counting used k=16, stride=1. Since each RNAseq read pair captured both strands of a cDNA molecule, each kmer and its reverse complement were counted together, using one canonical representation to represent both. The reads from each cross were processed separately. The kmer tabulation generated sets containing a count (positive integer) for each observed kmer (nucleotide sequence). The reads of each parent were processed separately. The kmer tabulation generated sets containing a count for each observed kmer. The parental kmer sets were filtered as follows. To exclude low-frequency kmers likely due to sequencing error, kmers with count<10 were filtered. To remove the numerous yet uninformative kmers shared by parents, the kmers in common were filtered from both parental sets, leaving their symmetric difference, for each parent pair. To remove the uninformative kmers absent from the cross, each parent set was filtered to retain only the kmers it had in common with the cross. Finally, the RNAseq pairs from each cross were compared to their parental kmer sets. Each pair was labeled as maternal origin if it contained at least one kmer found in the maternal-only kmer set and zero kmers found in the paternal-only kmer set. The opposite rule was used to label pairs of paternal origin. Most pairs could not be labeled.

Step 2. Assign labeled read pairs to gene: Reference transcripts were generated based on the *A. arenosa* reference (Bramsiepe 2023). Each reference used the given CDS sequence plus 150bp of upstream and downstream genomic sequence, added to help capture RNAseq read pairs that partially span the 5' or 3' UTR regions of the gene. Each read pair was mapped to at most one reference transcript, from which a gene ID was assigned. Custom software was used to count read pairs per gene per parent.

Step 3. Compute statistics: Counts of reads-per-gene were analyzed statistically to determine cases of allelic imbalance. In this step, the IRP procedure was used as described, without change. First, genes were filtered if any individual generated too few reads for statistical inference. Second, genes were filtered unless they demonstrated allelic balance in both parents of a cross. Third, genes were labeled as maternally or paternally biased if their maternal and paternal read count difference was statistically significant after correction for multiple tests.

Supplementary data resources

All scripts and SData files can be obtained at https:/github.com/PaulGrini/Hornslien_A_arenosa_imprinting. All sequences generated in this study have been deposited to the National Center for Biotechnology Information Sequence Read Archive (https://www.ncbi.nlm.nih.gov/sra) with project number PRJNAxxxxxx.

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